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New Enormous Classes of Non–Shannon Type Constrained Information Inequalities and their Relation to Finite Groups

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Abstract

Information inequalities govern the impossibilities in information theory and are therefore widely used for proving converse theorems in information theory. Most of the information inequalities for discrete random variables are consequences of the non-negativity of Shannon's information measures. Zhang and Yeung have recently derived a new information inequality for four discrete random variables and proved that their inequality implies $2^{14} - 1$ non-Shannon type constrained information inequalities. The existence of the so called non-Shannon type information inequalities indicates that the basic inequalities which stem from the non-negativity of Shannon's information measures do not fully characterize the joint entropy function of $n \geq 4$ discrete random variables. In this paper, we construct new classes of non-Shannon type constrained information inequalities whose cardinality grows *double exponentially* with n ($n \geq 4$). That construction generalizes the class of the Non-Shannon type constrained information inequalities which was reported by Yeung and Zhang for four discrete random variables. Relying on a fundamental relation between information theory and group theory which was devised by Chan and Yeung, we relate our discussion to finite groups.

1 Introduction

Shannon's information measures refer to entropies, conditional entropies, mutual information and conditional mutual information. It is well known that for discrete random variables these quantities are always non-negative (in contrast to differential entropies and conditional differential entropies of continuous random variables which may also attain negative values). Throughout the paper, we restrict our consideration to *discrete* random variables.
Following the notations in [2], let $\mathcal{N} = \{1, 2, \cdots, n\}$ and let $\Omega = \{X_i, i \in \mathcal{N}\}$ be any collection of n discrete random variables. For any subset α of $\mathcal{N}$, let $X_\alpha = \{X_i, i \in \alpha\}$ and $H_\Omega(\alpha) = H(X_\alpha)$. Associated with Ω are the $2^n - 1$ joint entropies: $H_\Omega(\alpha)$, $\forall\, \alpha \subseteq \mathcal{N}$ ($H_\Omega(\phi) = 0$, as the entropy of an empty set ϕ of random variables is zero). It is well known that for any Ω, $H_\Omega(\cdot)$ satisfies the following properties for all $\alpha, \beta \subseteq \mathcal{N}$:

1. $H_\Omega(\alpha) \leq H_\Omega(\beta)$, $\forall\, \alpha \subseteq \beta$,
2. $H_\Omega(\alpha) + H_\Omega(\beta) \geq H_\Omega(\alpha \cup \beta) + H_\Omega(\alpha \cap \beta)$.

These *basic inequalities* are called the *polymatriod axioms*, which are equivalent to the non-negativity of all the Shannon's information measures involving discrete random variables in Ω.

During the last decade, researchers have made significant progress in understanding the properties of Shannon's information measures (e.g, [1]–[5] and references therein). Recently, Zhang and Yeung have derived the following information inequality [[5], Theorem 3], which states that for any four discrete random variables:

$$I(X_3;X_4) \leq I(X_3;X_4|X_1) + I(X_3;X_4|X_2) + 0.5I(X_1;X_2) + 0.25I(X_1;X_3,X_4) + 0.25I(X_2;X_3,X_4) \quad (1)$$

which is also called the ZY98 inequality ([1], [2], [3]). Relying on (1), Zhang and Yeung have proved that the basic inequalities provide an *incomplete* characterization of the joint entropies of $n \geq 4$ discrete random variables [5]. Based on their observation, inequality (1) is called a non-Shannon type information inequality, since it cannot be proved as a consequence of the non-negativity of Shannon's information measures for arbitrary discrete random variables. Based on (1), Yeung and Zhang have also constructed a new class of $2^{14} - 1$ non-Shannon type constrained information inequalities for four discrete random variables ([2], [3]).

In this paper, we construct new classes of non-Shannon type constrained information inequalities for $n \geq 4$ discrete random variables. We show that the cardinality of these classes grows *double exponentially* with n. Relying on an intriguing and fundamental relation between information theory and group theory [1], we derive some non-trivial inequalities which are valid for finite groups.

2 New Non-Shannon Type Constrained Information Inequalities

Let $U, Z, X_1, X_2, \cdots, X_n$ be $n+2$ discrete random variables (where $n \geq 2$). Then, the following information inequality was derived by Zhang and Yeung (see [[5], Theorem 5]):

$$nI(U;Z) - 2\sum_{i=1}^{n} I(U;Z|X_i) \leq \frac{1}{n}\sum_{i=1}^{n} I(X_i;U,Z) + \sum_{i=1}^{n} H(X_i) - H(X_1,X_2,\cdots,X_n).$$

By replacing n with $n-2$ and then by substituting: $U = X_{n-1}, Z = X_n$, we obtain:

$$\begin{aligned}&(n-2)\, I(X_{n-1};X_n) + H(X_1,X_2,\cdots,X_{n-2}) \leq \\ &2\sum_{i=1}^{n-2} I(X_{n-1};X_n|X_i) + \left(\frac{1}{n-2}\right)\sum_{i=1}^{n-2} I(X_i;X_{n-1},X_n) + \sum_{i=1}^{n-2} H(X_i)\ ,\ \ n \geq 4.\end{aligned}$$

Based on the identity: $\sum_{i=1}^{n-2} H(X_i) - H(X_1,\cdots,X_{n-2}) = \sum_{i=2}^{n-2} I(X_1,\cdots,X_{i-1};X_i)$ for $n \geq 4$ (which follows from the chain rule for the entropy function), one obtains the following inequality:

$$\begin{aligned}I(X_{n-1};X_n) \leq\ & \frac{2}{n-2}\sum_{i=1}^{n-2} I(X_{n-1};X_n|X_i) + \frac{1}{n-2}\sum_{i=2}^{n-2} I(X_1,X_2,\cdots,X_{i-1};X_i) \\ &+ \frac{1}{(n-2)^2}\sum_{i=1}^{n-2} I(X_i;X_{n-1},X_n)\ ,\end{aligned} \quad (2)$$

We note that the information inequality (1) is a particular case of (2) (for the case where $n = 4$). Let $h_\alpha \triangleq H(X_\alpha)$, then it can be verified (see Appendix A) that inequality (2) can be expressed equivalently in the form:

$$\begin{aligned}&(n-2)^2\,(h_{n-1}+h_n) + (n-2)\,h_{1,2,\cdots,n-2} + (n-3)\,\textstyle\sum_{i=1}^{n-2} h_i \\ &+(2n-3)\textstyle\sum_{i=1}^{n-2} h_{i,n-1,n} \leq 2(n-2)\sum_{i=1}^{n-2}[h_{i,n-1}+h_{i,n}] + (n-2)(n-1)h_{n-1,n}\end{aligned} \qquad n \geq 4. \quad (3)$$

In the continuation of this section we will rely on the information inequality (3) and construct new classes of Non-Shannon type constrained information inequalities for $n \geq 4$ discrete random variables. The main idea here is to determine h_α for all $\alpha \subseteq \mathcal{N}$, such that the following basic inequalities are satisfied:

$$\begin{aligned} &h_\phi = 0, \\ &h_\alpha \leq h_\beta \ , \ \forall\, \alpha, \beta : \ \alpha \subseteq \beta \subseteq \mathcal{N}, \\ &h_\alpha + h_\beta \geq h_{\alpha \cup \beta} + h_{\alpha \cap \beta} \ , \ \forall\, \alpha, \beta \subseteq \mathcal{N}, \end{aligned} \tag{4}$$

but on the other hand inequality (3) *isn't* satisfied. That will imply that although $\{h_\alpha\}_{\alpha \subseteq \mathcal{N}}$ satisfies the basic inequalities, it cannot represent the joint entropies of any $n \geq 4$ discrete random variables, as otherwise inequality (3) which is known to be valid for every $n \geq 4$ discrete random variables would have been satisfied. By finding such $\{h_\alpha\}_{\alpha \subseteq \mathcal{N}}$ (which will be later called the h-vector), then it necessarily implies that (3) is a non-Shannon type information inequality. That means that the basic inequalities (4) are not sufficient for proving such an information inequality, as otherwise it should be satisfied for every h-vector which satisfies the basic inequalities. Then we will find as many as possible information measures which would have vanished *independently* if the selected h-vector could hypothetically represent the joint entropy functions of $n \geq 4$ discrete random variables (the independence requirement assures that there is no information measure within that set which is necessarily nulled as a consequence of nulling some other information measures in that set). Let $l(n)$ be the number of these independent constraints (where each constraint refers to nulling another information measure from the set above), then it is possible to construct a class of $2^{l(n)} - 1$ non-Shannon type constrained information inequalities for $n \geq 4$ discrete random variables. This can be done by combining the unconditional information inequality (2) (which in our notations is equivalent to (3)) with all the possible non-empty subsets of the above $l(n)$ independent constraints (this concept also appears in [2]).

In [2], a rather simple h-vector was suitable for constructing a class of non-Shannon type constrained information inequalities with four discrete random variables (its selection was based on inequality (1) and by following the approach which was outlined above):

$$\begin{aligned} &h_1 = h_2 = h_3 = h_4 = 2a, \quad a > 0, \\ &h_{1,2} = 4a, \ h_{1,3} = h_{1,4} = h_{2,3} = h_{2,4} = h_{3,4} = 3a, \\ &h_{1,2,3} = h_{1,2,4} = h_{1,3,4} = h_{2,3,4} = h_{1,2,3,4} = 4a. \end{aligned}$$

This h-vector is an appropriate selection for the particular case of $n = 4$, as it satisfies the basic inequalities (4) but doesn't satisfy the information inequality (1) (which is equivalent to (3) for the case where $n = 4$).

By attempting to generalize the above h-vector for the case where $n \geq 4$ such that the basic inequalities (4) will be satisfied and on the other hand inequality (3) will *not* be satisfied, we suggest (after some tedious trials) the following h-vector for $n \geq 4$:

$$\begin{aligned} &h_1 = h_2 = \cdots = h_n = (n-2)a, \\ &h_{n-1,n} = (2n-5)a, \\ &h_{i_1,i_2,\cdots i_m} = m(n-2)a, \\ &h_{i_1,i_2,\cdots i_m,n-1} = h_{i_1,i_2,\cdots i_m,n} = [(m+1)n - (3m+2)]a, \\ &h_{i_1,i_2,\cdots i_m,n-1,n} = [(m+2)n - 4(m+1)]a, \end{aligned} \qquad \begin{aligned} &a > 0, \ m \geq 1, \\ &1 \leq i_1 < \cdots < i_m \leq n-2, \end{aligned} \tag{5}$$

The considerations which led to the selection of the h-vector in (5) are presented in Appendix B. Let $f_n(h)$ be defined as the difference between the left and the right sides of inequality (3). Then,

for the h-vector in (5): $f_n(h) = (n-2)(n-3)a > 0$ (as $n \geq 4$ and $a > 0$), which implies that in spite of the fact that the h-vector in (5) satisfies the basic inequalities (4), it doesn't satisfy the information inequality (3) and therefore cannot represent the joint entropy functions $\{H(X_\alpha)\}_{\alpha \subseteq \mathcal{N}}$ of any $n \geq 4$ discrete random variables.

We list now the information measures which would have vanished independently if the selected h-vector in (5) could hypothetically represent the joint entropy functions of $n \geq 4$ discrete random variables (see Appendix B for more details):

$$\left.\begin{array}{l} I(X_i;X_j) = 0 \\ I(X_i;X_j|X_{n-1}) = 0 \\ I(X_i;X_j|X_n) = 0 \end{array}\right. \quad , \quad 1 \leq i < j \leq n-2\,, \tag{6}$$

$$\left.\begin{array}{l} I(X_i;X_{n-1}|X_n) = 0 \\ I(X_i;X_n|X_{n-1}) = 0 \end{array}\right. \quad , \quad 1 \leq i \leq n-2\,, \tag{7}$$

$$I(X_{n-1};X_n|X_{i_1},X_{i_2},\cdots,X_{i_m}) = 0, \quad \begin{array}{c} 1 \leq m \leq n-2 \\ 1 \leq i_1 < \cdots < i_m \leq n-2\,, \end{array} \tag{8}$$

$$\begin{array}{l} H(X_n|X_1,X_2,\cdots,X_{n-1}) = 0, \\ H(X_{n-1}|X_1,X_2,\cdots,X_{n-2},X_n) = 0. \end{array} \tag{9}$$

For the case where $n = 4$, one also obtains: $H(X_1|X_2,X_3,X_4) = 0$, $H(X_2|X_1,X_3,X_4) = 0$ (these two equalities are not satisfied for $n > 4$, and also a generalization of these equalities for $n > 4$ is not possible: $H(X_i|X_1,X_2,\cdots,X_{i-1},X_{i+1},\cdots,X_n) > 0$, $\forall 1 \leq i \leq n-2$). We note that according to (5), all the components of the h-vector are defined separately for every $n \geq 4$ (as they depend on n). In appendix C, we demonstrate that no information measures in (6)–(9) is necessarily nulled as a consequence of setting to zero a subset of the other information measures.

For $n \geq 4$, the total number of the constraints in (6)–(9) is:

$$\begin{aligned} l(n) &= \frac{3(n-2)(n-3)}{2} + 2(n-2) + (2^{n-2}-1) + [2 + 2\delta(n-4)] \\ &= 2^{n-2} + \frac{(3n-5)(n-2)}{2} + 1 + 2\delta(n-4), \end{aligned}$$

where the term $2\delta(n-4)$ (where $\delta(k)$ equals unity for $k = 0$ and is zero elsewhere) stands for the two additional constraints for the case where $n = 4$: $H(X_1|X_2,X_3,X_4) = H(X_2|X_1,X_3,X_4) = 0$ (these constraints are not satisfied if $n > 4$, and also in the later case:

$$H(X_i|X_1,X_2,\cdots,X_{i-1},X_{i+1},\cdots,X_n) > 0 \ , \ \forall\, 1 \leq i \leq n-2).$$

For $n \geq 4$, we construct a class of $2^{l(n)} - 1$ non-Shannon type constrained information inequalities with n discrete random variables. Every constrained information inequality of this class combines the unconditional information inequality (2) with an arbitrary non-empty subset of the above $l(n)$ independent constraints in (6)–(9). We conclude that for $n \geq 4$, the number of these non-Shannon type constrained information inequalities grows double exponentially with n (as $l(n)$ grows exponentially with n). For the particular case where $n = 4$, the $l(4) = 14$ constraints in (6)–(9) coincide with those indicated in [2], and also inequality (3) is equivalent to (1). Therefore, the new classes of the non-Shannon type constrained information inequalities which were constructed here

generalizes the class of the $2^{14}-1$ non-Shannon type information inequalities which was reported by Yeung and Zhang ([2], [3]).

We note here that the cardinality of an arbitrary class of non-Shannon type constrained information inequalities with n discrete random variables *cannot* grow more rapidly than double exponentially with n (where $n \geq 4$). The reason is that the set of the all the joint entropies of up to n discrete random variables has a cardinality of $2^n - 1$, and any set of independent constraints on Shannon's information measures (where all of these information measures are nulled) cannot have a cardinality which is above the cardinality of the former set. Since the cardinality of a class of non-Shannon type information inequalities is given by $2^{l(n)} - 1$ where $l(n) \leq 2^n - 1$, then the cardinality of every class of non-Shannon type constrained information inequalities which is constructed by combining a non-Shannon type un-constrained information inequality with an arbitrary subset of its related $l(n)$ independent constraints cannot grow more rapidly than double exponentially with n. Here we have constructed a class of non-Shannon type constrained information inequalities whose cardinality grows double exponentially with n, hence this asymptotic growth rate is indeed achievable.

3 Related Inequalities for Finite Groups

Chan and Yeung have found a one-to-one correspondence between group-theoretic inequalities and information inequalities [1]. Following their notations, let G be a finite group and let $G_1, G_2, \cdots, G_n$ be subgroups of G. A group-theoretic inequality was defined in [1] as an inequality which involves only additions and subtractions of terms of the form $\log\left(\frac{|G|}{|G_\alpha|}\right)$, where $|G|$ is the order of G and $|G_\alpha|$ is the order of the subgroup $\cap_{i\in\alpha} G_i$.
For any information inequality, one can establish a group-theoretic inequality through the following transformation: For any entropy term in the information inequality, say $h_\alpha = H(X_\alpha)$ (the entropy of the joint random variables $(X_i, i \in \alpha)$), we change it to $\log\left(\frac{|G|}{|G_\alpha|}\right)$. Then the inequality obtained is a group-theoretic inequality, which is valid for all finite groups. Chan and Yeung have demonstrated that a linear information inequality is valid if and only if its corresponding group-theoretic inequality is valid [1]. This theorem establishes an intriguing relation between entropy functions and finite groups in the form of inequalities. A trivial implication of the theorem is that if we can prove an information inequality, then we can also prove its corresponding group-theoretic inequality and vice versa.

Based on this fundamental relation which was stated in [1], we get from the information inequality (3) that if G is a finite group and $G_1, G_2, \cdots, G_n$ are subgroups of G (where $n \geq 4$), then the following inequality holds:

$$\begin{aligned}&\left(\prod_{i=1}^{n-2}|G_i\cap G_{n-1}|\,|G_i\cap G_n|\right)^{2(n-2)}\cdot|G_{n-1}\cap G_n|^{(n-2)(n-1)}\leq(|G_{n-1}||G_n|)^{(n-2)^2}\cdot\\&|G_1\cap G_2\cap\cdots\cap G_{n-2}|^{n-2}\cdot\left(\prod_{i=1}^{n-2}|G_i|\right)^{n-3}\cdot\left(\prod_{i=1}^{n-2}|G_i\cap G_{n-1}\cap G_n|\right)^{2n-3}\end{aligned}\quad,\quad n\geq 4. \tag{10}$$

By substituting $n = 4$ in inequality (10), we get as a particular case the group-theoretic inequality which was derived in [[1], Section IV].
Before we proceed, we list here some special cases where inequality (10) becomes trivial:
1. If $G_1 = G_2 = \cdots = G_n$ then (10) is satisfied with equality.

2. If $G_{n-1} = G_n = G$ then after some straightforward algebra we obtain from (10):

$$\prod_{i=1}^{n-2} |G_i| \leq |G_1 \cap G_2 \cap \cdots \cap G_{n-2}| \cdot |G|^{n-3} \quad , \quad n \geq 4.$$

However, the last inequality can be easily proved based on the inequality: $|G_1||G_2| \leq |G||G_1 \cap G_2|$ which is valid for any two subgroups of a finite group G and the property that if G_1 and G_2 are subgroups of G, then also $G_1 \cap G_2$ is a subgroup of G. The one-to-one correspondence which was mentioned above implies that the last inequality is equivalent to the trivial information inequality $H(X_1) + H(X_2) \geq H(X_1, X_2)$ (as was also noted in [1]).

3. If $G_1 \subseteq G_2 \subseteq \cdots \subseteq G_n$ then after some straightforward algebra we obtain from (10):

$$\prod_{i=1}^{n-2} |G_i| \leq \frac{|G_1||G_n|^{n-2}}{|G_{n-1}|} \quad , \quad n \geq 4.$$

The last inequality is trivial since in that case:

$$\frac{|G_1||G_n|^{n-2}}{|G_{n-1}|} \geq |G_1||G_{n-1}|^{n-3} = |G_1 \cap G_2 \cap \cdots \cap G_{n-2}| \cdot |G_{n-1}|^{n-3}.$$

But since $G_1, G_2, \cdots, G_{n-2}$ are subgroups of G_{n-1}, then from the discussion above we also obtain:

$$\prod_{i=1}^{n-2} |G_i| \leq |G_1 \cap G_2 \cap \cdots \cap G_{n-2}| \cdot |G_{n-1}|^{n-3}.$$

In the above particular cases, the non-trivial group-theoretic inequality (10) has led to three trivial inequalities. Inequality (10) is clearly *asymmetric* with respect to the indexes of the subgroups. In order to symmetrize that inequality with respect to the indexes of the $n \geq 4$ subgroups, we calculate the geometric means of the left and right sides of (10) with respect to all the $n!$ possible permutations of the indexes $1, 2, \cdots, n$. Since inequality (10) is satisfied for any individual permutation of the indexes, then it is clearly satisfied also for the corresponding geometric means over these permutations.

After some calculations (see Appendix D), we obtain the following group-theoretic inequality:

$$(P_{2,n})^{5n-9} \leq (P_{1,n})^{3n-7} \cdot (P_{3,n})^{2n-3} \cdot P_{n-2,n} \quad , \quad n \geq 4, \tag{11}$$

where:

$$P_{m,n} = \left\{ \prod_{1 \leq i_1 < i_2 < \cdots < i_m \leq n} |G_{i_1} \cap G_{i_2} \cap \cdots \cap G_{i_m}| \right\}^{\frac{1}{\binom{n}{m}}} \quad , \quad 1 \leq m \leq n. \tag{12}$$

As expected, (11) and (12) yield a symmetric inequality with respect to the n indexes of the subgroups $G_1, G_2, \cdots, G_n$. $P_{m,n}$ in (12) represents the geometric mean of the orders of all the subgroups which result in by the intersection of m out of the n subgroups $G_1, G_2, \cdots, G_n$.
It is rather simple to prove the inequality: $P_{1,n} \geq P_{2,n} \geq P_{3,n} \geq \cdots \geq P_{n,n}$ for all integer n. Therefore, for any $n \geq 4$: $P_{2,n} \geq P_{n-2,n}$. By invoking the last inequality, (11) can be loosened to the following inequality:

$$P_{2,n} \leq (P_{1,n})^{\beta_n} (P_{3,n})^{1-\beta_n} \quad , \beta_n \triangleq \frac{3n-7}{5n-10} \quad , \quad n \geq 4. \tag{13}$$

Clearly, $0.5 \leq \beta_n < 0.6$ for $n \geq 4$, and the group-theoretic inequality (13) yields a non-trivial relation between $P_{1,n}, P_{2,n}$ and $P_{3,n}$ for every finite group G and $n \geq 4$ subgroups $G_1, G_2, \cdots, G_n$ of G (in addition to the trivial relation where $P_{1,n} \geq P_{2,n} \geq P_{3,n}$). For the particular case where $n = 4$, (13) states that $P_{2,n}$ is upper bounded by the geometric mean of $P_{1,n}$ and $P_{3,n}$.

4 Summary and Conclusions

The existence of the so called non-Shannon type information inequalities is a consequence of the fact that the basic inequalities (4) which stem from the non-negativity of Shannon's information measures do not fully characterize the joint entropy function of $n \geq 4$ discrete random variables ([4], [5]). In this paper, we construct new classes of non-Shannon type constrained information inequalities for $n \geq 4$ discrete random variables (see section 2). Our construction is based on a variation of an information inequality which was originally derived by Zhang and Yeung [5]. It generalizes the class of the non-Shannon type constrained information inequalities which was recently reported by Yeung and Zhang for four discrete random variables ([2], [3]). We demonstrate that these new classes comprise non-Shannon type constrained information inequalities whose number grows *double exponentially* with n (e.g, the cardinalities of these classes are $2^{14}-1$, $2^{73}-1$, $2^{357}-1$ and $2^{2236}-1$ for $n = 4, 7, 10, 13$ discrete random variables, respectively). Relying on a fundamental relation between information theory and group theory which was recently devised by Chan and Yeung [1], we obtain non-trivial inequalities which are valid for $n \geq 4$ subgroups of an arbitrary finite group (see section 3).

The study here and also in [1]–[5] arise some fundamental questions which deserve further research:

- Is it possible to devise a systematic method for the derivation of non-Shannon type information inequalities for n discrete random variables (without constraints) ? Could the fundamental relation in [1] and the group theory help to devise such a method ?
- Do there exist non-Shannon type information inequalities for $n \geq 4$ discrete random variables (other then the inequality in section 2) which enable to construct classes of non-Shannon type constrained information inequalities whose cardinalities grow double exponentially with n (as was the situation here) ?
- How to determine systematically under which *independent* constraints a non-Shannon type information inequality stays in the same type ? (here it was specifically done for the non-Shannon type information inequality in section 2 by appropriately choosing the number of these independent constraints to be as large as possible).
- Possible applications of non-Shannon type information inequalities (such as for tightening outer bounds on the capacity regions of certain multi-terminal systems).

Appendix A: The equivalence between (2) and (3)

The equivalence between (2) and (3) stems from the following equalities:

$$\sum_{i=1}^{n-2} H(X_i) - H(X_1, X_2, \cdots, X_{n-2}) = \sum_{i=1}^{n-2} I(X_1, X_2, \cdots, X_{i-1}; X_i) \ , \ n \geq 4 \ ,$$

$$I(X_{n-1}; X_n | X_i) = H(X_i, X_{n-1}) + H(X_i, X_n) - H(X_i, X_{n-1}, X_n) - H(X_i) \ ,$$

$$I(X_i; X_{n-1}, X_n) = H(X_i) + H(X_{n-1}, X_n) - H(X_i, X_{n-1}, X_n) \ ,$$

$$I(X_{n-1}; X_n) = H(X_{n-1}) + H(X_n) - H(X_{n-1}, X_n) \ .$$

By substituting in (2) these four equalities and then by rearranging the terms in the left and right sides of (2), one obtains the following inequality for $n \geq 4$:

$$(n-2)^2\,(H(X_{n-1})+H(X_n))+(n-2)\,H(X_1,X_2,\cdots,X_{n-2})+$$
$$(n-3)\sum_{i=1}^{n-2}H(X_i)+(2n-3)\sum_{i=1}^{n-2}H(X_i,X_{n-1},X_n)\leq$$
$$2(n-2)\sum_{i=1}^{n-2}[H(X_i,X_{n-1})+H(X_i,X_n)]+(n-2)(n-1)H(X_{n-1},X_n)\,,$$

which coincides with (3) in our notations.

Appendix B: The Considerations for Selecting $\{h_\alpha\}_{\alpha\subseteq\mathcal{N}}$ as in (5)

We will try here to choose appropriately $\{h_\alpha\}_{\alpha\subseteq\mathcal{N}}$ for $n \geq 4$, such that our selection will generalize its selection in [2] (which was applied to the particular case where $n = 4$). In parallel, its selection should satisfy all the basic inequalities (4), but on the other hand should not satisfy (3) (the rational of these requirements was already explained in section 2).

As for the particular case where $n = 4$ (see [2]):

$$h_1 = h_2 = h_3 = h_4 = 2a, \quad a > 0\,,$$
$$h_{1,2} = 4a,\ h_{1,3} = h_{1,4} = h_{2,3} = h_{2,4} = h_{3,4} = 3a\,,$$

then for $n \geq 4$, we try (after some trials which were unfortunately not successful) to determine $\{h_\alpha\}_{\alpha\subseteq\mathcal{N}}$ by specifying some of its values:

$$\begin{aligned} &h_1 = h_2 = \cdots = h_n = (n-2)a, \\ &h_{i,j} = (2n-4)a, \quad 1 \leq i < j \leq n-2, \quad a > 0, \\ &h_{i,j} = (2n-5)a, \quad i < j \ \ , j = n-1, n, \end{aligned} \tag{B.1}$$

and then we will try to determine the other values of $\{h_\alpha\}_{\alpha\subseteq\mathcal{N}}$ according to the requirements which were mentioned above.

Based on the basic inequalities:

$$h_{i,n-1} + h_{j,n-1} \geq h_{i,j,n-1} + h_{n-1}\,, \quad 1 \leq i < j \leq n-2\,,$$

we obtain: $h_{i,j,n-1} \leq (3n-8)a$ for $1 \leq i < j \leq n-2$.
Moreover, for $1 \leq i < j \leq n-2$: $h_{i,j,n-1} \geq \max\{h_{i,j}, h_{i,n-1}, h_{i,n}, h_{j,n-1}, h_{j,n}\} = (2n-4)a$.
For $1 \leq i < j \leq n-2$ and $n \geq 4$, we therefore obtain: $(2n-4)a \leq h_{i,j,n-1} \leq (3n-8)a$, and similarly $(2n-4)a \leq h_{i,j,n} \leq (3n-8)a$.
In order to determine later $h_{1,2,\cdots,n-2}$ to be as large as possible (as to increase the value of the left side of (3), which gives a chance that (3) will not be satisfied by the selected h-vector), we determine $h_{i,j,n-1}$ and $h_{i,j,n}$ to be the maximal value within the permissible range of values:

$$h_{i,j,n-1} = h_{i,j,n} = (3n-8)a\,, \quad 1 \leq i < j \leq n-2. \tag{B.2}$$

For $1 \leq i \leq n-2$, we obtain the basic inequality:

$$h_{i,n-1} + h_{i,n} \geq h_{i,n-1,n} + h_i\,,$$

which yields: $h_{i,n-1,n} \leq (3n-8)a$ $(\forall\, 1 \leq i \leq n-2)$.
On the other hand, based on the basic inequalities (4):

$$h_{i,n-1,n} \geq \max\{h_{i,n-1}, h_{i,n}, h_{n-1,n}\} = (2n-5)a.$$

From the same reason which was mentioned above, we determine $h_{i,n-1,n}$ to be the maximal possible value:

$$h_{i,n-1,n} = (3n-8)a, \quad 1 \leq i \leq n-2. \tag{B.3}$$

Based on the basic inequality:

$$h_{i,n-1,n} + h_{j,n-1,n} \geq h_{i,j,n-1,n} + h_{n-1,n}\,, \quad 1 \leq i < j \leq n-2\,,$$

we obtain from (B.1) and (B.3): $h_{i,j,n-1,n} \leq (4n-11)a$ (for $1 \leq i < j \leq n-2$). Moreover, based on the basic inequalities, we get for $n \geq 4$:

$$h_{i,j,n-1,n} \geq \max\{h_{i,n-1,n}, h_{j,n-1,n}, h_{i,j,n-1}, h_{i,j,n}\} = (3n-8)a\,, \quad 1 \leq i < j \leq n-2\,.$$

In order to generalize the following constraint that was achieved in [2] for the case where $n = 4$:

$$I(X_3; X_4|X_1) = I(X_3; X_4|X_2) = I(X_3; X_4|X_1, X_2) = 0\,,$$

we will select in the continuation of this appendix the values of the h-vector such that constraint (8) will be satisfied for $n \geq 4$, which is equivalent to the following equality:

$$h_{i_1,i_2,\cdots,i_m,n-1,n} = h_{i_1,i_2,\cdots,i_m,n-1} + h_{i_1,i_2,\cdots,i_m,n} - h_{i_1,i_2,\cdots,i_m}\,, \qquad \begin{array}{c} n \geq 4\,, \\ 1 \leq i_1 < \cdots < i_m \leq n-2\,. \end{array} \tag{B.4}$$

In particular, from equalities (B.2) and (B.4) we obtain:

$$h_{i,j,n-1,n} = h_{i,j,n-1} + h_{i,j,n} - h_{i,j} = (4n-12)a\,, \quad 1 \leq i < j \leq n-2\,.$$

Since we obtained the inequality $(3n-8)a \leq h_{i,j,n-1,n} \leq (4n-11)a$, then it is indeed possible to make the choice:

$$h_{i,j,n-1,n} = (4n-12)a\,, \quad 1 \leq i < j \leq n-2\,. \tag{B.5}$$

It can be easily verified that for the particular case where $n = 4$, the choice of the components of the h-vector so far coincides with [2] (as indicated in section 2).

For $1 \leq i_1 < i_2 < i_3 \leq n-2$, we get from the basic inequalities:

$$h_{i_1,i_2,i_3,n-1} + h_{i_1,n-1} \leq h_{i_1,i_2,n-1} + h_{i_1,i_3,n-1}\,,$$

and based on (B.1), (B.2), we obtain: $h_{i_1,i_2,i_3,n-1} \leq (4n-11)a$ and similarly $h_{i_1,i_2,i_3,n} \leq (4n-11)a$. Moreover, $h_{i_1,i_2,i_3} \leq h_{i_1,i_3} + h_{i_2,i_3} - h_{i_3} = (3n-6)a$.
In order to determine in the continuation a value of $h_{1,2,\cdots,n-2}$ which will be as large as possible (as was explained before), we determine:

$$\begin{array}{l} h_{i_1,i_2,i_3,n-1} = h_{i_1,i_2,i_3,n} = (4n-11)a \\ h_{i_1,i_2,i_3} = (3n-6)a \end{array}\,, \quad 1 \leq i_1 < i_2 < i_3 \leq n-2. \tag{B.6}$$

We note that since there are in (B.6) three integers i_1, i_2, i_3 such that $1 \leq i_1 < i_2 < i_3 \leq n-2$, then it implies that in this case $n \geq 5$, which insures that $(4n-11)a = h_{i_1,i_2,i_3,n-1} \geq h_{i_1,i_2,i_3} = (3n-6)a$ $(a > 0)$, as could be expected from the basic inequalities.
From (B.4) and (B.6), we get that for $1 \leq i_1 < i_2 < i_3 \leq n-2$:

$$h_{i_1,i_2,i_3,n-1,n} = h_{i_1,i_2,i_3,n-1} + h_{i_1,i_2,i_3,n} - h_{i_1,i_2,i_3} = (5n-16)a.$$

By continuing this process, it can be easily verified by mathematical induction that for $n \geq 4$, the resulting choice for $\{h_\alpha\}_{\alpha \subseteq \mathcal{N}}$ coincides with (5). In particular, we obtain from (5): $h_{1,2,\cdots,n-2} = (n-2)^2 a$.

So far we have determined $\{h_\alpha\}_{\alpha \subseteq \mathcal{N}}$ to satisfy all the basic inequalities (4) for $n \geq 4$, and in addition it satisfies the equality (B.4). Let $f_n(h)$ be a function of the h-vector which designates the difference between the left and right sides of inequality (3):

$$\begin{aligned} f_n(h) = {} & (n-2)^2 (h_{n-1} + h_n) + (n-2)\, h_{1,2,\cdots,n-2} + (n-3) \textstyle\sum_{i=1}^{n-2} h_i + \\ & (2n-3) \textstyle\sum_{i=1}^{n-2} h_{i,n-1,n} - 2(n-2) \sum_{i=1}^{n-2} [h_{i,n-1} + h_{i,n}] - (n-2)(n-1) h_{n-1,n} \end{aligned} \quad , \quad n \geq 4. \tag{B.7}$$

Then, for the selected h-vector in (5) we get: $f_n(h) = (n-2)(n-3)a > 0$ (as $n \geq 4$ and $a > 0$). The selection of the h-vector in (5) satisfies all the basic inequalities, but do not satisfy inequality (3), and therefore it cannot represent the joint entropy of any $n \geq 4$ discrete random variables.

If $\{h_\alpha\}_{\alpha \subseteq \mathcal{N}}$ could hypothetically represent the joint entropy of $n \geq 4$ discrete random variables, then all the information measures in (6)–(9) would have vanished. To verify it, we express all these information measures as linear combinations of unconditional joint entropies:

$$\begin{aligned} I(X_i; X_j) &= H(X_i) + H(X_j) - H(X_i, X_j) = h_i + h_j - h_{i,j} \,, \\ I(X_i; X_j | X_{n-1}) &= H(X_i, X_{n-1}) + H(X_j, X_{n-1}) - H(X_{n-1}) - H(X_i, X_j, X_{n-1}) \\ &= h_{i,n-1} + h_{j,n-1} - h_{n-1} - h_{i,j,n-1} \,, \\ I(X_i; X_j | X_n) &= H(X_i, X_n) + H(X_j, X_n) - H(X_n) - H(X_i, X_j, X_n) \\ &= h_{i,n} + h_{j,n} - h_n - h_{i,j,n} \,, \end{aligned}$$

and for $1 \leq i < j \leq n-2$ and $n \geq 4$, the selection of the h-vector in (5) implies (6).
For $1 \leq i \leq n-2$ we get:

$$\begin{aligned} I(X_i; X_{n-1} | X_n) &= H(X_i, X_n) + H(X_{n-1}, X_n) - H(X_i, X_{n-1}, X_n) - H(X_n) \\ &= h_{i,n} + h_{n-1,n} - h_{i,n-1,n} - h_n \,, \\ I(X_i; X_n | X_{n-1}) &= H(X_i, X_{n-1}) + H(X_{n-1}, X_n) - H(X_i, X_{n-1}, X_n) - H(X_{n-1}) \\ &= h_{i,n-1} + h_{n-1,n} - h_{i,n-1,n} - h_{n-1} \,, \end{aligned}$$

and the selection of the h-vector in (5) implies (7) for $n \geq 4$.
Based on equality (B.4), we get the property:

$$\begin{aligned} & I(X_{n-1}; X_n | X_{i_1}, X_{i_2}, \cdots, X_{i_m}) = \\ & H(X_{i_1}, X_{i_2}, \cdots, X_{i_m}, X_{n-1}) + H(X_{i_1}, X_{i_2}, \cdots, X_{i_m}, X_n) - \\ & H(X_{i_1}, X_{i_2}, \cdots, X_{i_m}) - H(X_{i_1}, X_{i_2}, \cdots, X_{i_m}, X_{n-1}, X_n) = \\ & h_{i_1,i_2,\cdots,i_m,n-1} + h_{i_1,i_2,\cdots,i_m,n} - h_{i_1,i_2,\cdots,i_m,n-1,n} - h_{i_1,i_2,\cdots,i_m} = 0 \,, \end{aligned} \quad \begin{aligned} & n \geq 4 \,, \\ & 1 \leq i_1 < \cdots < i_m \leq n-2 \,, \end{aligned} \tag{B.8}$$

which verifies (8) for $n \geq 4$.
Finally, for the selection of the h-vector in (5):

$$\begin{aligned} H(X_n | X_1, X_2, \cdots, X_{n-1}) &= H(X_1, X_2, \cdots, X_{n-1}, X_n) - H(X_1, X_2, \cdots, X_{n-1}) \\ &= h_{1,2,\cdots,n} - h_{1,2,\cdots,n-1} = 0, \end{aligned}$$

$$\begin{aligned} H(X_{n-1} | X_1, X_2, \cdots, X_{n-2}, X_n) &= H(X_1, X_2, \cdots, X_{n-1}, X_n) - H(X_1, X_2, \cdots, X_{n-2}, X_n) \\ &= h_{1,2,\cdots,n} - h_{1,2,\cdots,n-2,n} = 0, \end{aligned}$$

which verifies (9).
For the particular case where $n = 4$, one also obtains the two equalities:

$$\begin{aligned} H(X_1|X_2,X_3,X_4) &= H(X_1,X_2,X_3,X_4) - H(X_2,X_3,X_4) \\ &= h_{1,2,3,4} - h_{2,3,4} = 0, \\ H(X_2|X_1,X_3,X_4) &= H(X_1,X_2,X_3,X_4) - H(X_1,X_3,X_4) \\ &= h_{1,2,3,4} - h_{1,3,4} = 0. \end{aligned}$$

Appendix C: On the Independence of the Constraints (6)–(9)

In this appendix, we wish to demonstrate that there are no information measures in (6)–(9) which are nulled as a consequence of setting to zero a subset of the other information measures.

First we will show that the constraints in (6) and the other constraints in (7)–(9) are independent: Suppose that X_i and X_j are statistically independent for some pairs of (i,j) where $1 \leq i < j \leq n-2$ and in addition they may be also statistically independent when the values of the random variables X_{n-1} and X_n are known (i.e, let us assume that all or part of the constraints in (6) are satisfied). These constraints may be satisfied even if there exists an integer k $(1 \leq k \leq n-2)$ such that $X_n = X_k$ when the value of the random variable X_{n-1} is known. This situation implies that the inequality $I(X_k; X_n|X_{n-1}) > 0$ could be satisfied for every $1 \leq k \leq n-2$. Similarly, if we replace $n-1$ with n, then also the inequality $I(X_k; X_{n-1}|X_n) > 0$ could be satisfied for $1 \leq k \leq n-2$. In the trivial case where $X_{n-1} = X_n$, all the information measures in (8) are positive if and only if $H(X_n) > 0$. In the case where $X_1, X_2, \cdots, X_n$ are statistically independent (which then implies that all the constraints which are expressed in (6) are satisfied), then for any $1 \leq i \leq n$ we obtain $H(X_i|X_1, \cdots, X_{i-1}, X_{i+1}, \cdots, X_n) = H(X_i) > 0$, unless these random variables are degenerate. Therefore, if all the constraints in (6) or a subset of these constraints are satisfied, there is no constraint in (7)–(9) which also should be satisfied.
On the other hand, if all or a subset of the constraints in (7)–(9) are satisfied, then clearly there might be a statistical dependence between X_i and X_j for every pair (i,j) such that $1 \leq i < j \leq n-2$ (and also there might be such a statistical dependence conditioning on the random variables X_n, X_{n-1}). That situation implies that no information measure in (6) should necessarily vanish in case that all or part of the information measures in (7)–(9) are nulled. Therefore, we conclude that setting to zero any subset of the information measures in (6) doesn't necessarily imply that some information measures in (7)–(9) should vanish and vice versa.

It is clear that the constraints in (6) are independent (in general, conditioning may increase or decrease the mutual information between two random variables). Moreover, the constraints in (7)–(9) are independent. To see this, suppose X_{n-1} and X_n are statistically independent in case that the values of a specific set (or some specific sets) of the $n-2$ random variables $X_1, X_2, \cdots, X_{n-2}$ are known (which means that a subset of the information measures in (8) are set to zero). Then, the knowledge of the values of any other set of these $n-2$ random variables may of course introduce a certain level of statistical dependence between the random variables X_{n-1} and X_n, which then implies that all the other conditional mutual information in (8) between these two random variables may be strictly positive. Therefore, in that case, all the other constraints in (8) may not be satisfied. Moreover, the case where X_{n-1} and X_n are statistically independent if the values of specific subsets of the other $n-2$ random variables are known, does not avoid of course the possibility of a statistical dependence between every other pair of these n random variables. For examples, the random variables X_k and X_{n-1} may be statistically dependent either if the value of X_n is known or unknown, and similarly a statistical dependence may exist between the random variables X_k and

X_n for some $1 \leq k \leq n-2$ either if the value of the random variable X_{n-1} is known or not. That will also imply that the information measures which appear in (7) and (9) will be strictly positive. Moreover, if $X_{n-1} = X_i$ for $1 \leq i \leq n-2$ and the random variables X_{n-1} and X_n are statistically independent no matter how many values of the random variables $X_1, X_2, \cdots, X_{n-2}$ are known, then $I(X_i; X_{n-1}|X_n) > 0$, but $I(X_i; X_n|X_{n-1})$ is nulled and also the constraints in (8) and (9) are satisfied (which means that the all the information measures in (8) and (9) are set to zero). Similarly, if we switch between n and $n-1$, then we get a situation where $I(X_i; X_n|X_{n-1}) > 0$, but all the other constraints in (7)–(9) are satisfied. Finally, if for an integer number k (where $1 \leq k \leq n$), the conditional entropy $H(X_k|X_1, X_2, \cdots, X_{k-1}, X_{k+1}, \cdots, X_n)$ is set to zero (which also encompass the two additional constraints for $n = 4$ in section 2), then a statistical dependence between every pair of the random variables may still exist, which yields that all the information measures in (6)–(8) may attain positive values. It also doesn't imply that for *some* other integer $1 \leq j \leq n$, the random variable X_j can be determined if the values of the other random variables are known (which means that all the other information measures in (9) may also attain positive values). From the discussion above, we conclude that the constraints in (6)–(9) are independent.

Appendix D: Derivation of (11), (12)

In this appendix, we will start our discussion by presenting a theorem that apart of enabling us to simplify the derivation of (11), (12), it is also more general then the case where we utilize it here. First, we will start our discussion by introducing an operator:

Definition: Let $f = f(u_1, u_2, \cdots, u_n)$ be a function of n variables.
We define T_n to be an operator which calculates the geometric mean of all the $n!$ values of the function f which are calculated for all the permutations of the coordinates $(u_1, u_2, \cdots, u_n)$:

$$T_n(f) \triangleq \left\{ \prod_{\pi} f\left(u_{\pi(1)}, u_{\pi(2)}, \cdots, u_{\pi(n)}\right) \right\}^{\frac{1}{n!}}, \tag{D.1}$$

where the product in (D.1) is over all the $n!$ permutations π of $\mathcal{N}$.

Example 1: Let $f(u_1, u_2, u_3) = u_1$, then:

$$T_3(f) = (u_1^2 u_2^2 u_3^2)^{\frac{1}{3!}} = (u_1 u_2 u_3)^{\frac{1}{3}}$$

The following theorem enables to simplify the calculation of $T_n(f)$ under some symmetry conditions:

Theorem:

1. If f, g are functions of the same set of n variables, then: $T_n(fg) = T_n(f)\, T_n(g)$.

2. If the function $f(u_1, u_2, \cdots, u_n)$ depends on m out of its n variables (where $1 \leq m \leq n$), i.e there exists a function g which satisfies the equality:

$$f(u_1, u_2, \cdots, u_n) = g(u_{i_1}, u_{i_2}, \cdots, u_{i_m})$$

for some m indexes $i_1, i_2, \cdots, i_m$ (where $1 \leq i_1 < i_2 < \cdots < i_m \leq n$), and if the value of the function g is invariant to permutations of its m variables, then the following equality holds:

$$T_n(f) = \left(\prod_{1 \leq j_1 < j_2 < \cdots < j_m \leq n} g(u_{j_1}, u_{j_2}, \cdots, u_{j_m}) \right)^{\frac{1}{\binom{n}{m}}}. \tag{D.2}$$

Proof:
Part 1 of the theorem is trivial, so we will concentrate on proving the second part of the theorem. Under our assumptions, it follows that $f\left(u_{\pi(1)}, u_{\pi(2)}, \cdots, u_{\pi(n)}\right)$ depends only on the m variables whose indexes are $\pi(i_1), \pi(i_2), \cdots, \pi(i_m)$, and:

$$f\left(u_{\pi(1)}, u_{\pi(2)}, \cdots, u_{\pi(n)}\right) = g\left(u_{\pi(i_1)}, u_{\pi(i_2)}, \cdots, u_{\pi(i_m)}\right) ,$$

for every permutation π of $\mathcal{N}$.
Since the function g is invariant to permutations of its m coordinates then there are $m!\,(n-m)!$ permutations of $\mathcal{N}$, such that the value of $f\left(u_{\pi(1)}, u_{\pi(2)}, \cdots, u_{\pi(n)}\right)$ stays constant. Therefore, one can partition the set of the $n!$ possible permutations of $\mathcal{N}$ to $\binom{n}{m}$ subsets of $m!\,(n-m)!$ permutations, where for each subset of these permutations the value of $f\left(u_{\pi(1)}, u_{\pi(2)}, \cdots, u_{\pi(n)}\right)$ stays constant. For every subset of these permutations, lets choose its representative as the unique permutation for which $1 \le \pi(i_1) < \pi(i_2) < \cdots < \pi(i_m) \le n$. Going through all the $\binom{n}{m}$ subsets of permutations, we get as their representative permutations the set of *all* the permutations of $\mathcal{N}$ which satisfy the inequality: $1 \le \pi(i_1) < \pi(i_2) < \cdots < \pi(i_m) \le n$.
Let $j_k = \pi(i_k)$ (where $k = 1, 2, \cdots, m$), then we get from (D.1):

$$\begin{aligned} T_n(f) &= \left(\prod_{1 \le j_1 < j_2 < \cdots < j_m \le n} g(u_{j_1}, u_{j_2}, \cdots, u_{j_m})^{m!(n-m)!} \right)^{\frac{1}{n!}} \\ &= \left(\prod_{1 \le j_1 < j_2 < \cdots < j_m \le n} g(u_{j_1}, u_{j_2}, \cdots, u_{j_m}) \right)^{\frac{1}{\binom{n}{m}}} , \end{aligned}$$

which completes our proof.

Example 2: If $f(u_1, u_2, u_3) = u_1 u_2^2 (u_1 + u_2)^2 (u_1 + u_3)^4$, then based on the theorem above:

$$\begin{aligned} T_3(f) &= T_3(u_1)\, T_3(u_2^2)\, T_3((u_1+u_2)^2)\, T_3((u_1+u_3)^4) \\ &= T_3(u_1)\, \left(T_3(u_2)\right)^2\, \left(T_3(u_1+u_2)\right)^2\, \left(T_3(u_1+u_3)\right)^4 \\ &= (u_1u_2u_3)^{\frac{1}{3}}\, (u_1u_2u_3)^{\frac{2}{3}}\, \left((u_1+u_2)(u_1+u_3)(u_2+u_3)\right)^{\frac{2}{3}}\, \left((u_1+u_2)(u_1+u_3)(u_2+u_3)\right)^{\frac{4}{3}} \\ &= u_1u_2u_3\, (u_1+u_2)^2 (u_1+u_3)^2 (u_2+u_3)^2. \end{aligned}$$

We turn now to the derivation of (11), (12). Since (10) is valid for every permutation of $\mathcal{N}$, then it implies that for $n \ge 4$:

$$T_n\left(\left(\prod_{i=1}^{n-2} |G_i \cap G_{n-1}|\, |G_i \cap G_n| \right)^{2(n-2)} \cdot |G_{n-1} \cap G_n|^{(n-2)(n-1)} \right) \le$$

$$T_n\left(\left(|G_{n-1}||G_n|\right)^{(n-2)^2} \cdot |G_1 \cap G_2 \cap \cdots \cap G_{n-2}|^{n-2} \cdot \left(\prod_{i=1}^{n-2} |G_i| \right)^{n-3} \cdot \left(\prod_{i=1}^{n-2} |G_i \cap G_{n-1} \cap G_n| \right)^{2n-3} \right) .$$

Now, based on the theorem which was presented above, we obtain the following equalities for the

left side of the last inequality:

$$T_n\left(\left(\prod_{i=1}^{n-2}|G_i\cap G_{n-1}|\,|G_i\cap G_n|\right)^{2(n-2)}\cdot|G_{n-1}\cap G_n|^{(n-2)(n-1)}\right)=$$

$$\left(T_n\left(\prod_{i=1}^{n-2}|G_i\cap G_{n-1}|\,|G_i\cap G_n|\right)\right)^{2(n-2)}\cdot\left(\,T_n\left(|G_{n-1}\cap G_n|\right)\,\right)^{(n-2)(n-1)}=$$

$$\left(\prod_{i=1}^{n-2}T_n(|G_i\cap G_{n-1}|)\cdot T_n(|G_i\cap G_n|)\right)^{2(n-2)}\cdot\left(\,T_n\left(|G_{n-1}\cap G_n|\right)\,\right)^{(n-2)(n-1)}=$$

$$\left(\prod_{1\le j_1<j_2\le n}|G_{j_1}\cap G_{j_2}|^{\frac{2}{n(n-1)}}\right)^{4(n-2)^2}\cdot\left(\prod_{1\le j_1<j_2\le n}|G_{j_1}\cap G_{j_2}|^{\frac{2}{n(n-1)}}\right)^{(n-2)(n-1)}=$$

$$\left(\prod_{1\le j_1<j_2\le n}|G_{j_1}\cap G_{j_2}|\right)^{\frac{2(5n-9)(n-2)}{n(n-1)}}.$$

Similarly, the calculation of the right side of the last inequality yields:

$$\left(\prod_{i=1}^{n}|G_i|\right)^{\frac{(n-2)(3n-7)}{n}}\cdot\left(\prod_{1\le j_1<j_2<j_3\le n}|G_{j_1}\cap G_{j_2}\cap G_{j_3}|\right)^{\frac{6(2n-3)}{n(n-1)}}\cdot$$

$$\left(\prod_{1\le j_1<j_2<\cdots<j_{n-2}\le n}|G_{j_1}\cap G_{j_2}\cap\cdots\cap G_{j_{n-2}}|\right)^{\frac{2(n-2)}{n(n-1)}}.$$

The substitution of these two results in the left and right sides of the last inequality yields (11) and (12).

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